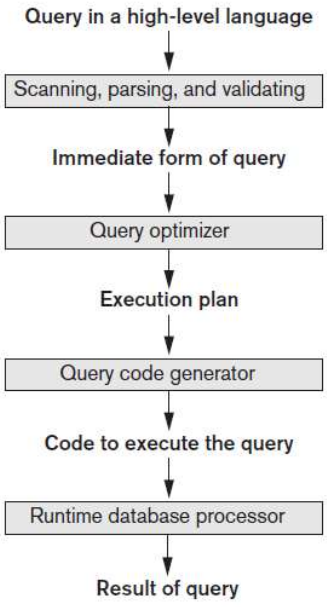
Chapter 11

Query Optimization

1

Query Processing



In SQLITE:

```
EXPLAIN QUERY PLAN
<query>
```

2

Complexity of Natural Join

assume index on key

Join On	$\max \lvert R_1(n) \bowtie R_2(m) \rvert$	Runtime complexity
Key(R_1) – Key(R_2)	$\min(m, n)$	$O(n \log m)$ (assume $n \le m$)
Key(R_1) – NonKey(R_2)	m	$O(m \log n)$
NonKey(R_1) – Key(R_2)	n	$O(n \log m)$
NonKey(R_1) – NonKey(R_2)	mn	$O(nm)$

Theorem: In the natural join $R_1 \bowtie R_2 \bowtie \ldots \bowtie R_k$, such that $R_i \bowtie R_{i+1}$ is on a unique attribute of R_{i+1}, we have:

$$|R_1 \bowtie R_2 \bowtie \ldots \bowtie R_k| \le |R_1|$$

Conclusion: Since join is commutative and associative - for efficiency, it is best to order a sequence of natural joins in increasing order of table size.

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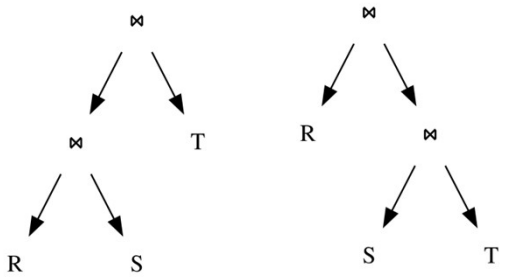
Size Matters

Table	#Rows	Runtime
R	10	
S	10^4	
T	10^6	
$R \bowtie S$	10	$40 \log 10$
$S \bowtie T$	10^4	$60{,}000 \log 10$
$(R \bowtie S) \bowtie T$	10	$40 \log 10 + 60 \log 10 = \mathbf{100 \log 10}$
$R \bowtie (S \bowtie T)$	10	$60{,}000 \log 10 + 40 \log 10 = 60{,}040 \log 10$

$R \bowtie S \bowtie T$



$(R \bowtie S) \bowtie T$ left deep

$R \bowtie (S \bowtie T)$ right deep

4

Use Relational Algebra

List all employees who earn more than any employee in dept #5.

```
SELECT Lname, Fname
FROM   EMPLOYEE
WHERE Salary > (SELECT MAX(Salary)
                FROM EMPLOYEE
                WHERE Dno=5);
```

Inner block

```
( SELECT MAX (Salary)
  FROM   EMPLOYEE
  WHERE  Dno=5 )
```

$\Im_{\text{MAX Salary}}(\sigma_{\text{Dno=5}}(\text{EMPLOYEE}))$

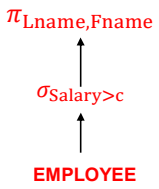
Outer block

```
SELECT  Lname, Fname
FROM    EMPLOYEE
WHERE Salary > c
```

$\pi_{\text{Lname,Fname}}(\sigma_{\text{Salary>c}}(\text{EMPLOYEE}))$

The query optimizer chooses an execution plan for each query block.

This is easy since the inner block is *uncorrelated* with the outer block.



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Query Trees

List the names and titles of employees working on a project whose budget is at least $250,000 and which employs more than two employees.

```
SELECT      E.Ename, E.Title
FROM        EMPLOYEE E, PROJECT P, WORKS_ON W
WHERE       P.Budget > 250000
      AND   E.Ssn = W.Essn
      AND   P.Pnumber = W.Pno
      AND   P.Pnumber IN
                  (SELECT W.Pno
                   FROM   WORKS_ON W
                   GROUP BY W.Pno
                   HAVING COUNT(*) > 2)
```

How should we execute this query?

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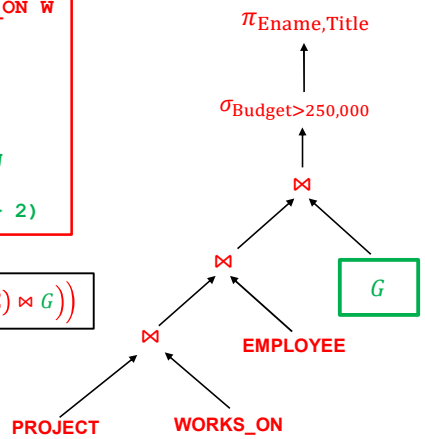
Query Trees

```
SELECT  E.Ename, E.Title
FROM            EMPLOYEE E, PROJECT P, WORKS_ON W
WHERE           P.Budget > 250000
        AND     E.Ssn = W.Essn
        AND     P.Pnumber = W.Pno
        AND     P.Pnumber IN
                        (SELECT W.Pno
                         FROM   WORKS_ON W
                         GROUP BY W.Pno
                         HAVING COUNT(*) > 2)
```

$$\pi_{Ename,Title}\left(\sigma_{Budget>250,000}\left(((\text{PROJECT} \bowtie \text{WORKS\_ON}) \bowtie \text{EMPLOYEE}) \bowtie G\right)\right)$$



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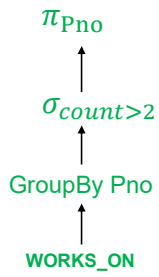
Query Trees

```
SELECT W.Pno
FROM   WORKS_ON W
GROUP BY W.Pno
HAVING COUNT(*) > 2
```

$$\pi_{Pno}(\sigma_{count>2} Pno\ \mathcal{F}\ \text{WORKS\_ON})$$



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Query Trees

```
SELECT  E.Ename, E.Title
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        AND     P.Pnumber = W.Pno
        AND     P.Pnumber IN
                        (SELECT W.Pno
                         FROM   WORKS_ON W
                         GROUP BY W.Pno
                         HAVING COUNT(*) > 2)
```

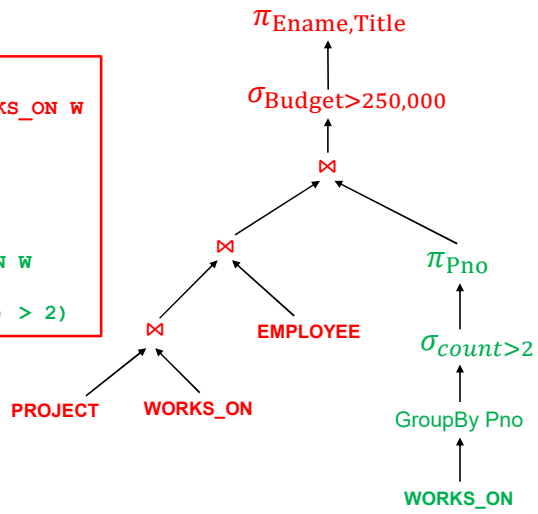

$\pi_{Ename,Title}\left(\sigma_{Budget>25\ ,000}\left(((\text{PROJECT} \bowtie \text{WORKS\_ON}) \bowtie \text{EMPLOYEE}) \bowtie \pi_{Pno}(\sigma_{count>2} Pno\mathcal{F}\text{WORKS\_ON})\right)\right)$

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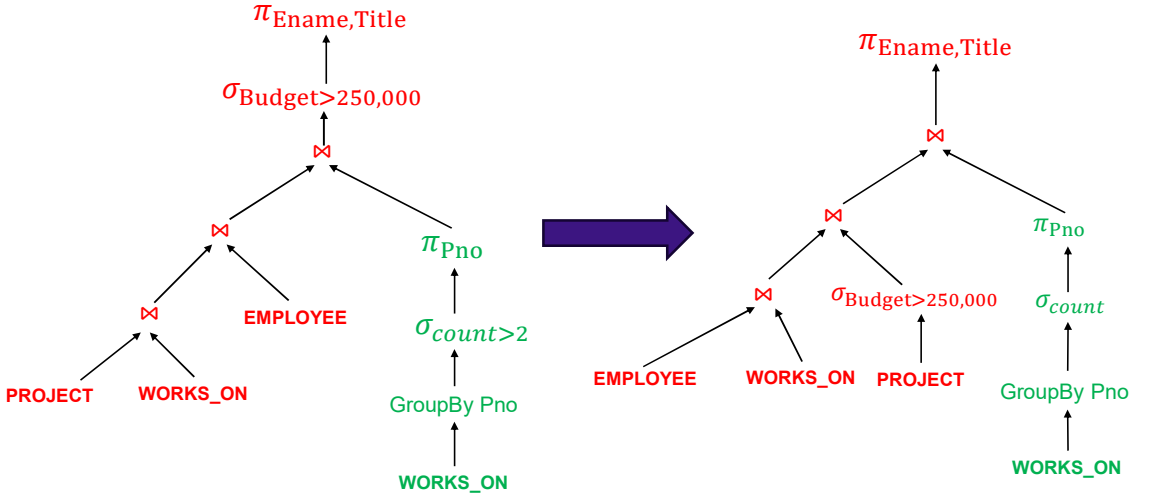
Alternative Query Tree



$\pi_{Ename,Title}\left(((\text{EMPLOYEE} \bowtie \text{WORKS\_ON}) \bowtie \sigma_{Budget>250,000}\ \text{PROJECT}) \bowtie \pi_{Pno}(\sigma_{count>2} Pno\mathcal{F}\text{WORKS\_ON})\right)$

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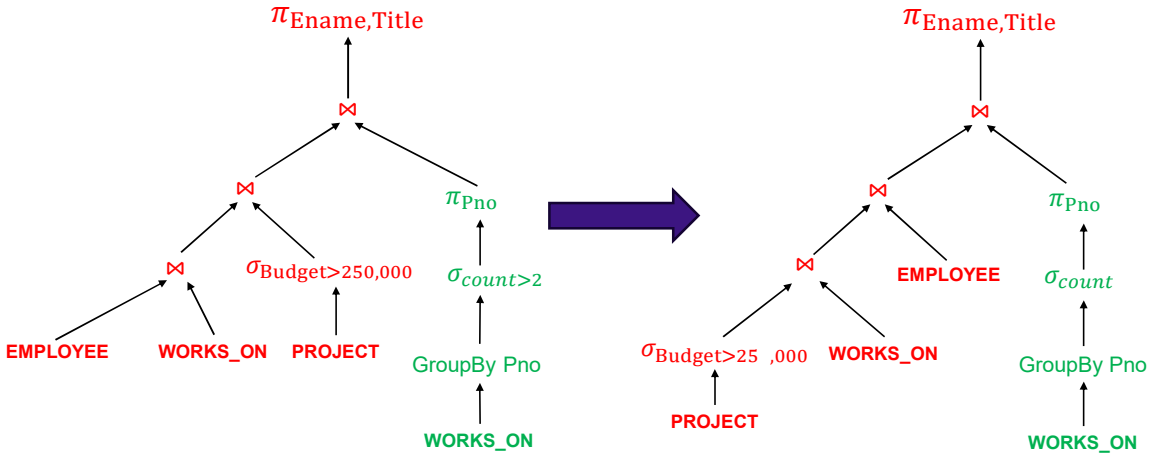
Alternative Query Tree



$$\pi_{Ename,Title}\left(\left(\left(\left(\sigma_{Budget>250,000}\text{PROJECT}\right)\bowtie \text{WORKS\_ON}\right)\bowtie \text{EMPLOYEE}\right)\bowtie \pi_{Pno}\left(\sigma_{count>2}\,Pno\mathcal{F}\text{WORKS\_ON}\right)\right)$$

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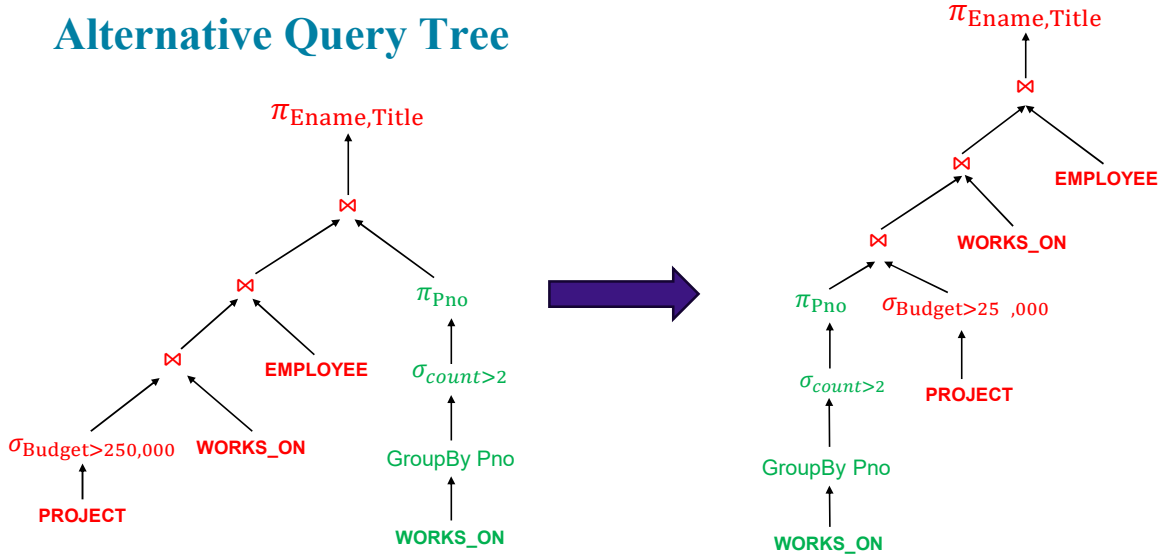
Alternative Query Tree



$$\pi_{Ename,Title}\left(\left(\left(\pi_{Pno}\left(\sigma_{count}\quad Pno\mathcal{F}\text{WORKS}_{\text{ON}}\right)\bowtie \sigma_{Budget>250,000}\ \text{PROJECT}\right)\bowtie \text{WORKS}_{\text{ON}}\right)\bowtie \text{EMPLOYEE}\right)$$

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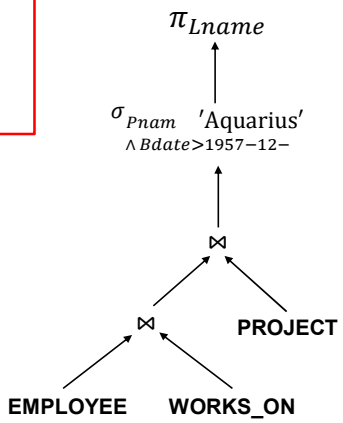
Query Transformation

```
SELECT      E.Lname
FROM        EMPLOYEE E, PROJECT P, WORKS_ON W
WHERE       P.Pname = 'Aquarius'
      AND   E.Ssn = W.Essn
      AND   P.Pnumber = W.Pno
      AND   P.Bdate > 1957-12-31;
```



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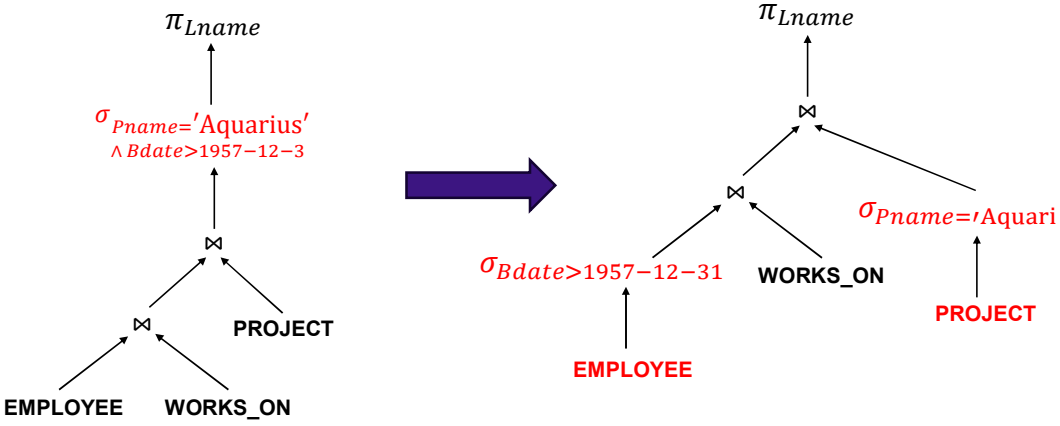
Query Transformation

Move SELECT operations down the query tree to reduce table sizes.



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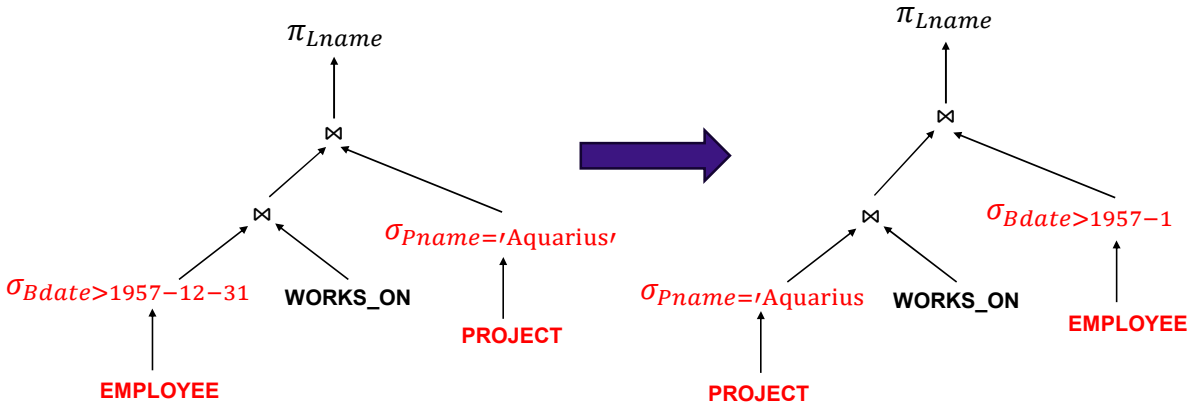
Query Transformation

Apply the more restrictive SELECT operation first to reduce join complexity.



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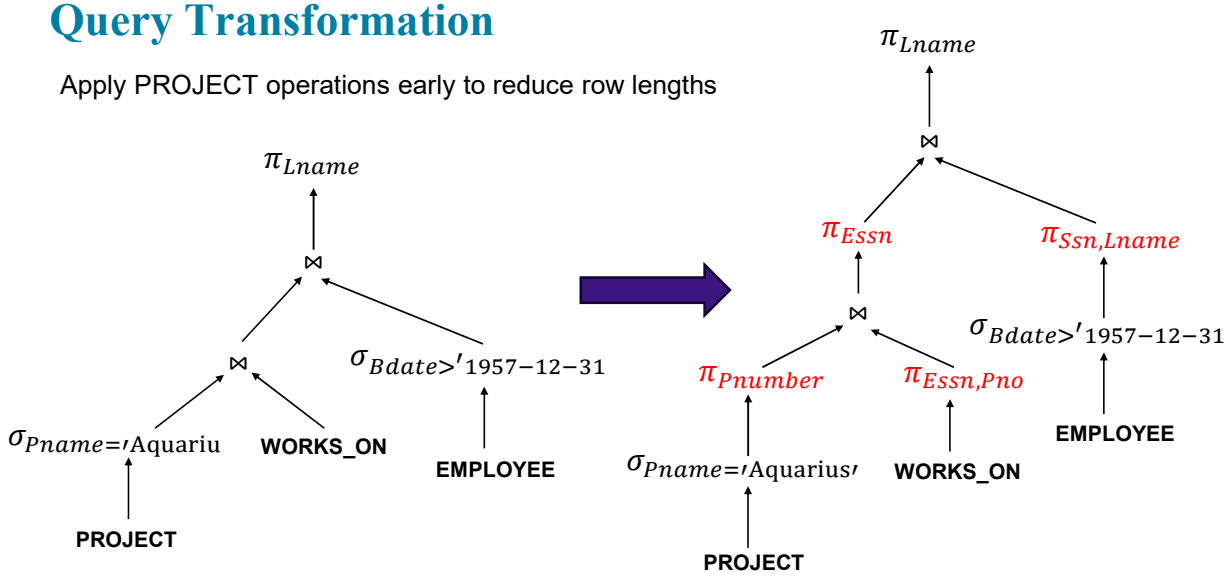
Query Transformation

Apply PROJECT operations early to reduce row lengths



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Transformation Rules for Rational Algebra Expressions

Cascade of σ:

- A conjunctive selection condition can be broken up into a cascade (sequence) of individual selections

$$\sigma_{A \wedge B}(R) = \sigma_A(\sigma_B(R))$$

Commutativity of σ:

$$\sigma_B(\sigma_A(R)) = \sigma_A(\sigma_B(R))$$

Cascade of π:

- In a cascade (sequence) of projections, all but the last one can be ignored

$$\pi_A(\pi_B(R)) = \pi_A(R) \quad A \subseteq B$$

Commutativity of σ with π:

$$\sigma_A(\pi_B(R)) = \pi_B(\sigma_A(R))$$

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Transformation Rules for Rational Algebra Expressions

Commutativity of $\bowtie$:

$$R \bowtie S = S \bowtie R$$

$\Rightarrow \quad \bowtie R_i = \bowtie R_{perm(i)}$

Associativity of $\bowtie$:

$$(R \bowtie S) \bowtie T = R \bowtie (S \bowtie T)$$

Distribution of σ over $\bowtie$:

$$\sigma_{C_1 \wedge C_2}(R \bowtie S) = \sigma_{C_1}(R) \bowtie \sigma_{C_2}(S)$$

if C_1 involves only Schema(R) and C_2 involves only Schema(S)

Distribution of π over $\bowtie$:

$$\pi_{A \cup B}(R \bowtie S) = \pi_{A \cup J}(R) \bowtie \pi_{B \cup J}(S)$$

if $A \subseteq$ Schema(R) and $B \subseteq$ Schema(S) and J are the join attributes

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Heuristics for Query Optimization

SELECT → PROJECT → JOIN

Apply first the operations that reduce the size of intermediate results

- Perform SELECT and PROJECT operations as early as possible to reduce the number of tuples and attributes
- The SELECT and JOIN operations that are most restrictive should be executed first
- Use indexes to speed up JOIN where possible

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